

Superposition Theorem

Statement

Superposition theorem states that:

In a linear circuit containing multiple independent sources, the voltage across or current through any element is the algebraic sum of the voltages or currents produced by each independent source acting alone.

Important Theory Terms

- Works only for **linear circuits**
 - Consider **one independent source at a time**
 - Other voltage sources → **short circuit**
 - Other current sources → **open circuit**
 - Final answer = **algebraic sum** of all individual effects
-

MCQ with Answer

1. Superposition theorem is applicable to—

- A) Non-linear circuit
- B) Linear circuit
- C) Digital circuit
- D) Open circuit

 **Answer: B) Linear circuit**

2. While applying superposition theorem, an ideal voltage source is replaced by—

- A) Open circuit
- B) Resistance
- C) Short circuit
- D) Current source

 **Answer: C) Short circuit**

3. While applying superposition theorem, an ideal current source is replaced by—

- A) Short circuit
- B) Open circuit
- C) Battery
- D) Inductor

 **Answer: B) Open circuit**

4. Superposition theorem is used to calculate—

- A) Power only
- B) Resistance only
- C) Voltage or current
- D) Frequency

 **Answer: C) Voltage or current**

Source Transformation Theorem

Statement

Source transformation theorem states that:

A voltage source in series with a resistor can be transformed into an equivalent current source in parallel with the same resistor and vice versa.

Important Theory Terms

- Voltage source + series resistance
↔ Current source + parallel resistance
 - Equivalent circuits produce same output
 - Used to simplify circuit analysis
-

MCQ with Answer

1. A voltage source with series resistance can be converted into—

- A) Capacitor
- B) Inductor

- C) Current source with parallel resistance
- D) Open circuit

✓ **Answer: C) Current source with parallel resistance**

2. In source transformation, the resistance value—

- A) Changes completely
- B) Becomes zero
- C) Remains same
- D) Doubles

✓ **Answer: C) Remains same**

3. Source transformation is mainly used for—

- A) Amplification
- B) Simplifying circuits
- C) Measuring frequency
- D) Power generation

✓ **Answer: B) Simplifying circuits**

4. Current source equivalent of a voltage source is connected with resistor in—

- A) Series
- B) Parallel
- C) Delta
- D) Star

✓ **Answer: B) Parallel**

Thevenin's Theorem

Statement

Thevenin's theorem states that:

Any linear bilateral network containing voltage sources, current sources, and resistances can be replaced by an equivalent circuit consisting of a single voltage source (V_{th}) in series with a resistance (R_{th}).

Important Theory Terms

- Applicable to **linear bilateral network**
 - Equivalent circuit contains:
 - Thevenin voltage (V_{th})
 - Thevenin resistance (R_{th})
 - Used to simplify complex networks
 - Load resistance is usually removed first
-

MCQ with Answer

1. Thevenin equivalent circuit consists of—

- A) Current source only
- B) Voltage source only
- C) Voltage source and series resistance
- D) Parallel resistance only

✓ **Answer: C) Voltage source and series resistance**

2. Thevenin theorem is applicable to—

- A) Linear bilateral circuits
- B) Non-linear circuits
- C) Magnetic circuits
- D) Mechanical systems

✓ **Answer: A) Linear bilateral circuits**

3. Thevenin resistance is represented by—

- A) (R_L)
- B) (R_{th})
- C) (R_s)
- D) (R_m)

✓ **Answer: B) (R_{th})**

4. In Thevenin theorem, the load resistor is—

- A) Shorted
- B) Removed
- C) Doubled
- D) Grounded

✓ **Answer: B) Removed**

Maximum Power Transfer Theorem

Statement

Maximum power transfer theorem states that:

Maximum power is transferred from source to load when the load resistance is equal to the internal resistance (Thevenin resistance) of the source network.

Important Theory Terms

- Condition:

$$R_L = R_{th}$$

- Maximum power occurs when:
 - Load resistance = source resistance
 - Commonly used in communication systems and electronics
-

MCQ with Answer

1. Maximum power transfer occurs when—

- A) ($R_L > R_{th}$)
- B) ($R_L < R_{th}$)
- C) ($R_L = R_{th}$)
- D) ($R_L = 0$)

✓ **Answer: C) ($R_L = R_{th}$)**

2. Maximum power transfer theorem is mainly used in—

- A) Communication systems
- B) Water pumps
- C) Mechanical gears
- D) Thermodynamics

☒ **Answer: A) Communication systems**

3. For maximum power transfer, load resistance should be—

- A) Infinite
- B) Zero
- C) Equal to source resistance
- D) Half of source resistance

☒ **Answer: C) Equal to source resistance**

4. Maximum power transfer theorem deals with—

- A) Voltage only
- B) Current only
- C) Power delivered to load
- D) Frequency response

☒ **Answer: C) Power delivered to load**

AC Related Important Theory

নিচে AC chapter এর important statement, theory terms, formula আর MCQ with answer গুছিয়ে দিলাম। এগুলো তোমার AC Problems file আর slide এর সাথে মিল রেখে করা হয়েছে।

Alternating Current (AC)

Statement

Alternating Current (AC) is an electric current whose magnitude and direction change periodically with time.

Important Theory Terms

- AC changes direction continuously
- সাধারণ waveform = sinusoidal wave
- AC voltage expression:

$$v(t) = V_m \cos(\omega t + \phi)$$

Where,

- (V_m) = maximum voltage
 - (ω) = angular frequency
 - (ϕ) = phase angle
-

MCQ with Answer

1. AC stands for—

- A) Active Current
- B) Alternating Current
- C) Alternate Charge
- D) Active Circuit

 **Answer: B) Alternating Current**

2. In AC circuit, current—

- A) Remains constant
- B) Changes direction periodically
- C) Flows one direction only
- D) Stops completely

✓ **Answer: B) Changes direction periodically**

Angular Frequency

Formula

genui {`"math_block_widget_always_prefetch_v2": {"content": "\omega = 2\pi f"}}`

Where,

- ω = angular frequency (rad/s)
 - f = frequency (Hz)
-

MCQ

1. Unit of angular frequency is—

- A) Volt
- B) Hertz
- C) Radian/second
- D) Ohm

✓ **Answer: C) Radian/second**

Inductive Reactance

Statement

Opposition offered by an inductor to alternating current is called inductive reactance.

Formula

$$X_L = \omega L$$

Important Terms

- Depends on frequency
 - Higher frequency $\rightarrow$ higher (X_L)
 - Unit = ohm (Ω)
-

MCQ

1. Inductive reactance increases with—

- A) Resistance
- B) Frequency
- C) Current
- D) Power

☒ **Answer: B) Frequency**

2. Unit of inductive reactance—

- A) Henry
- B) Farad
- C) Volt
- D) Ohm

☒ **Answer: D) Ohm**

Capacitive Reactance

Statement

Opposition offered by a capacitor to alternating current is called capacitive reactance.

Formula

$$X_C = \frac{1}{\omega C}$$

Important Terms

- Higher frequency $\rightarrow$ lower (X_C)
 - Unit = ohm (Ω)
-

MCQ

1. Capacitive reactance decreases with—

- A) Voltage
- B) Frequency
- C) Resistance
- D) Power

 **Answer: B) Frequency**

2. Capacitive reactance is inversely proportional to—

- A) Frequency
- B) Resistance
- C) Power
- D) Voltage

 **Answer: A) Frequency**

Impedance

Statement

Total opposition offered by an AC circuit to current flow is called impedance.

Formula

Series RLC impedance:

$$Z=R+j(X_L-X_C)$$

Magnitude:

$$|Z|=\sqrt{R^2+(X_L-X_C)^2}$$

Important Terms

- Symbol = (Z)
 - Unit = ohm (Ω)
 - Combination of resistance and reactance
-

MCQ

1. Unit of impedance—

- A) Watt
- B) Volt
- C) Ohm
- D) Coulomb

✓ Answer: C) Ohm

2. Impedance in AC circuit is represented by—

- A) R
- B) Z
- C) X
- D) V

✓ Answer: B) Z

Resonance

Statement

Resonance occurs in an RLC circuit when inductive reactance equals capacitive reactance.

Resonance Condition

$$X_L = X_C$$

Resonance Frequency Formula

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

Important Terms

- At resonance:
 - Impedance minimum
 - Current maximum
 - Power factor becomes unity
-

MCQ

1. Resonance occurs when—

- A) ($R = X_L$)
- B) ($X_L = X_C$)
- C) ($R = X_C$)
- D) ($X_L = 0$)

✓ Answer: B) ($X_L = X_C$)

2. At resonance, current becomes—

- A) Minimum
- B) Zero
- C) Maximum
- D) Constant

✓ Answer: C) Maximum

Phase Difference

Statement

The angular difference between voltage and current waveforms is called phase difference.

Important Terms

- In inductive circuit:
 - Current lags voltage
 - In capacitive circuit:
 - Current leads voltage
-

MCQ

1. In purely inductive circuit—

- A) Current leads voltage
- B) Voltage leads current
- C) Both equal
- D) No current flows

 **Answer: B) Voltage leads current**

2. In purely capacitive circuit—

- A) Current leads voltage
- B) Voltage leads current
- C) Both are equal
- D) Frequency becomes zero

 **Answer: A) Current leads voltage**

AC Circuit Current Formula

Formula

$$I = \frac{V}{Z}$$

MCQ

1. Current in AC circuit depends on—

- A) Voltage only
- B) Impedance only
- C) Voltage and impedance
- D) Resistance only

 **Answer: C) Voltage and impedance**

Leads & Lags

Important Theory

- If voltage phase > current phase
→ Voltage leads current
- If current phase > voltage phase
→ Current leads voltage

Example from your sheet:

$$\begin{aligned} &[\\ &30^\circ - 25.43^\circ = 4.57^\circ \\ &] \end{aligned}$$

Therefore,

- Voltage leads current by (4.57°)
- Current lags voltage by (4.57°)